

TryHackMe — Computer Fundamentals Module

Cloud Computing Fundamentals

Learning Notes — Service Models, Deployment Models, Cloud Providers & AWS EC2 Lab

"Imagine you have a fantastic idea for an app — but what do you do when users from across the world experience lag, or thousands connect at once? That is when cloud computing comes to play."



1. What Is Cloud Computing?

Cloud computing is the delivery of computing services — servers, storage, databases, networking, software, and analytics — over the internet ("the cloud") on a pay-as-you-go basis. Instead of owning and maintaining physical hardware, organisations and individuals rent access to computing resources from cloud providers, using only what they need and paying only for what they use.

The Origin Story — The App Problem

Imagine you have a fantastic app idea that helps students practice cyber security. You host it on your own computer. But what happens when users from around the world experience lag? What if thousands connect at once? What if your computer is turned off? These limitations are exactly what cloud computing solves — by providing scalable, globally distributed, always-available computing infrastructure on demand.

1.1 Cloud Computing Is Built on What You Already Know

The cloud is not magic — it is built on the same technologies covered in the previous Computer Fundamentals rooms:

Foundation Technology	How Cloud Uses It
Virtualisation	Cloud providers run thousands of physical servers, using Type 1 hypervisors to create virtual machines that are rented out to customers as IaaS.
Containers	Cloud platforms (AWS ECS, Google Kubernetes Engine) run containerised workloads at massive scale — faster and more efficiently than VMs for many use cases.
Networking	Cloud infrastructure is built on global fibre networks, DNS, HTTP(S), load balancers, and firewalls — all the networking concepts you have studied.
Storage	Cloud storage (S3, Azure Blob, Google Cloud Storage) uses distributed SSD/HDD storage arrays with redundancy and global replication.
Operating Systems	Cloud VMs run standard Linux and Windows Server instances — the same OS fundamentals apply in the cloud as on local machines.

1.2 Key Benefits of Cloud Computing

Benefit	What It Means
Scalability	Handle unexpected traffic spikes by automatically adding resources — scale up when demand rises, scale down when it falls. The room's key answer: scalability handles unexpected increases.
Cost Efficiency	Pay-as-you-go: no large upfront hardware investment. Pay only for what you use, when you use it. No idle hardware costs.
Global Reach	Deploy your application to data centres on every continent — users worldwide get low latency by connecting to the nearest region.
High Availability	Cloud providers offer SLA-guaranteed uptime (99.9%+) with automatic failover, redundant power, and multiple data centres.
Flexibility	Instantly provision new servers, databases, or storage — change infrastructure in minutes rather than weeks of hardware procurement.
Managed Services	Offload infrastructure management to the provider — focus on your application, not on patching servers or replacing failed drives.
Disaster Recovery	Cloud provides built-in backup, replication, and failover across geographic regions — protecting against localised outages.

Room Answer: What characteristic handles unexpected traffic increases?

Scalability — the ability to automatically increase (scale out) or decrease (scale in) computing resources in response to demand. This is the defining advantage of cloud over on-premises infrastructure.

2. Cloud Service Models — IaaS, PaaS, and SaaS

Cloud services are offered in three main models, each representing a different level of responsibility sharing between the provider and the customer. The room uses a rental analogy — different ways of renting a place to live — to make these models concrete and memorable.

The Rental Analogy

Think of cloud service models like different ways of renting a place to live: IaaS is renting an empty flat — you get the walls and plumbing, you furnish it yourself. PaaS is a furnished flat — furniture is provided, you bring your belongings. SaaS is a hotel — everything is provided, you just show up and use it.

IaaS — Infrastructure as a Service — Rent the raw infrastructure — servers, storage, networking

Rental Analogy: Renting an empty flat — you get walls, plumbing, electricity. You furnish and decorate it yourself.

Provider manages: Physical hardware, virtualisation, networking, storage hardware, data centre facilities.

You manage: Operating system, runtime, middleware, applications, data — everything above the hardware.

Examples: AWS EC2, Microsoft Azure VMs, Google Compute Engine, DigitalOcean Droplets.

Security note: You have full OS access and full responsibility for OS patching, security configuration, and vulnerability management. Misconfigured security groups (firewall rules) on EC2 instances are a leading cause of cloud breaches.

PaaS — Platform as a Service — Rent the platform — focus on building your app, not managing servers

Rental Analogy: Renting a furnished flat — furniture is already there. You bring your clothes and personal items.

Provider manages: Physical hardware, virtualisation, networking, OS, runtime, middleware, security patches.

You manage: Application code and data only — no server management required.

Examples: AWS Elastic Beanstalk, Heroku, Google App Engine, Azure App Service, Render.

Security note: Reduced attack surface (no OS management) but less visibility and control. Shared responsibility model shifts more to the provider — but you still own application security: input validation, authentication, authorisation, and data protection.



SaaS — Software as a Service — Use software over the internet — provider manages everything

Rental Analogy: *Staying in a hotel — everything is provided. You just show up and use the room.*

Provider manages: Everything: hardware, OS, runtime, application, maintenance, updates, and infrastructure security.

You manage: Your data and user access settings only — no infrastructure or application management.

Examples: Gmail, Google Workspace, Microsoft 365, Zoom, Salesforce, Slack, Dropbox.

Security note: Minimal technical responsibility but data governance is critical — your data is stored on provider infrastructure. Data privacy, compliance (GDPR), access controls, and account security (MFA) are your key responsibilities.

2.1 Responsibility Matrix — What Each Model Manages

Layer	IaaS	PaaS	SaaS
Applications	You	You	Provider
Data	You	You	Shared
Runtime	You	Provider	Provider
Middleware	You	Provider	Provider
Operating System	You	Provider	Provider
Virtualisation	Provider	Provider	Provider
Servers	Provider	Provider	Provider
Storage	Provider	Provider	Provider
Networking	Provider	Provider	Provider

Room Answer: Deploy an app focusing only on development, leaving infrastructure to others — which model?

PaaS (Platform as a Service) — the provider manages the infrastructure and OS, so you can focus exclusively on building and deploying your application.

3. Cloud Deployment Models

Cloud environments are not one-size-fits-all. Organisations choose between three deployment models based on their security requirements, regulatory obligations, budget, and how much control they need over their infrastructure.



Public Cloud — Shared infrastructure managed entirely by a third-party provider

- Infrastructure is owned and operated by a cloud provider (AWS, Azure, GCP) and shared among many organisations.
- Accessible over the public internet — any authorised user can connect from anywhere.
- Provider is responsible for physical security, hardware maintenance, and infrastructure reliability.
- Most cost-effective option — shared infrastructure means lower per-user cost.
- Virtually unlimited scalability — providers have data centres globally with massive capacity.
- Use cases: web applications, SaaS products, development environments, startups.
- Examples: AWS, Microsoft Azure, Google Cloud Platform, IBM Cloud, Oracle Cloud.

Security note: Shared infrastructure means other organisations' workloads run on the same physical hardware (though isolated by virtualisation). Compliance and data sovereignty concerns for regulated industries. Misconfigured resources (open S3 buckets, permissive security groups) are a leading cause of cloud data breaches.



Private Cloud — Dedicated cloud infrastructure for a single organisation

- Infrastructure is used exclusively by one organisation — either on-premises or hosted by a third-party as a dedicated environment.
- Maximum control, customisation, and privacy — the organisation owns and manages the environment.
- Required for highly regulated industries: banking, healthcare, defence, government — where data cannot leave controlled premises.
- Higher cost — no sharing of infrastructure means the organisation bears the full cost of hardware, maintenance, and staffing.
- Limited scalability compared to public cloud — bounded by owned hardware capacity.
- Can be built using the same technologies as public cloud (VMware, OpenStack, Nutanix).
- Use cases: financial institutions, government agencies, healthcare systems, defence contractors.

Security note: Private cloud provides the strongest security and compliance posture — data never leaves controlled premises. However, the organisation is fully responsible for all security controls, patching, and incident response. Internal threats and misconfiguration are the primary risks.



Hybrid Cloud — Combination of public and private cloud — best of both worlds

- A hybrid cloud combines private cloud (or on-premises) infrastructure with public cloud resources, connected securely.
- Sensitive data and regulated workloads stay in the private environment; scalable and less-sensitive workloads use public cloud.
- Cloud bursting: when private infrastructure reaches capacity, overflow workloads automatically move to public cloud.
- Enables gradual cloud migration — move workloads to public cloud incrementally rather than all at once.

- Requires secure, reliable connectivity between private and public environments (VPN, dedicated links like AWS Direct Connect).
- More complex to manage than either model alone — requires expertise in both environments.
- Use cases: enterprises balancing legacy systems with modern cloud services, regulated industries needing flexibility.

Security note: Hybrid environments increase attack surface — the connection between private and public cloud is a potential attack path. VPN tunnels and dedicated links between environments must be carefully secured. Identity management across both environments is complex — consistent access control policies are critical.

3.1 Deployment Model Comparison

Feature	Public	Private	Hybrid
Ownership	Third-party provider	Organisation-owned	Both
Cost	Low (pay-as-you-go)	High (full infra cost)	Medium
Scalability	Virtually unlimited	Limited by owned hardware	Flexible
Security control	Provider-managed baseline	Maximum — full control	Mixed
Compliance	Varies by provider cert	Highest — full data custody	Configurable
Management	Minimal (provider handles)	Full responsibility	Complex — both
Best for	Startups, web apps, dev	Banking, healthcare, defence	Enterprises, migration

4. Major Cloud Providers

There are several cloud vendors offering a wide variety of services. Understanding the major providers — their strengths and market positions — is important for both cloud architecture decisions and security assessments.

Provider	Market Position	Key Strengths & Notes
AWS	Industry Leader	Amazon Web Services is the largest cloud provider globally with the most extensive service catalogue and the broadest geographic coverage. Industry standard for cloud infrastructure. Used in the room's practical lab (EC2). Key services: EC2 (VMs), S3 (storage), RDS (databases), Lambda (serverless), VPC (networking).

Azure	Enterprise Leader	Microsoft Azure is a strong competitor especially in enterprise and hybrid cloud environments. Deep integration with Microsoft products (Windows Server, Active Directory, Office 365). Popular in enterprises already using Microsoft technology. Key services: Azure VMs, Azure AD, Azure Blob Storage, Azure DevOps.
GCP	Data & AI Leader	Google Cloud Platform is known for powerful data analytics, AI and machine learning tools. Runs on the same infrastructure as Google Search, YouTube, and Gmail. Popular for big data (BigQuery), ML (Vertex AI), and Kubernetes (GKE — Google originated Kubernetes). Key services: Compute Engine, BigQuery, GKE, Cloud Storage.

5. Practical Lab — AWS EC2 Deployment

The room concludes with a hands-on lab where you act as a cloud architect deploying a cyber security training application for students. You work with a simulated AWS console to provision three EC2 (Elastic Compute Cloud) virtual machines — practising Infrastructure as a Service (IaaS) in action.

5.1 Key EC2 Concepts

Concept	Explanation
EC2	Amazon Elastic Compute Cloud — virtual computers (VMs) in the AWS cloud. Each EC2 instance is a VM you can create, start, stop, resize, and terminate on demand.
Instance	A single running EC2 virtual machine. 'Adding an EC2 instance' = adding a virtual computer to your environment.
Instance Type	Defines the hardware specifications of the VM — how many vCPUs and how much RAM. Determines performance and cost. E.g. t2.micro (1 vCPU, 1GB RAM) vs m5.large (2 vCPU, 8GB RAM).
AMI	Amazon Machine Image — the template used to launch an EC2 instance. Contains the OS, pre-installed software, and configuration. Like a VM image or container image.
Region	Geographic location of the data centre where the instance runs. Choosing a region close to users reduces latency.
Security Group	A virtual firewall controlling inbound and outbound traffic to EC2 instances — which ports and protocols are allowed from which sources.

5.2 Instance Type Tiers — Choosing the Right Size

Instance types are organised in families and sizes. The room introduces three tiers relevant to the lab scenario:

Tier	vCPU / RAM	Cost	Best For
t2 (Burstable)	1–8 vCPU / 0.5–32GB	Low	Dev/test environments, low-traffic apps, learning labs — burstable CPU performance.
t3 (Burstable+)	2–8 vCPU / 0.5–32GB	Medium	General-purpose workloads — improved baseline performance over t2, same burst model.
m5 (General)	2–96 vCPU / 8–384GB	Higher	Production workloads requiring consistent CPU/memory performance without bursting.

5.3 Lab Tasks — Deploying the Cyber Security Training Platform

The lab simulates deploying a cyber security training app for students. You create three EC2 instances — representing the IaaS model where you manage the OS and application layer above the cloud infrastructure:

#	VM Name	Instance Type	Purpose
1	web-server-1	t2.micro	Frontend web server — serves the training application UI to students.
2	app-server-1	t3.medium	Application server — runs the backend logic and APIs for the training platform.
3	db-server-1	m5.large	Database server — stores student data, progress records, and training content.

Why IaaS for Security Training?

Cyber security practices often require full access to the operating system — to install tools, configure the system, and safely simulate attacks in an isolated environment. This is why the training platform uses IaaS (EC2 instances) rather than PaaS or SaaS. Full OS control is essential for security testing labs, which is also exactly how TryHackMe itself works.

6. Room Answers — Quick Reference

Question	Answer
What cloud characteristic handles unexpected access increases?	scalability
Deploy an app focusing only on development, leaving infra to others — model?	PaaS
Gmail and Zoom are examples of which cloud service model?	SaaS
AWS EC2 = which cloud service model?	IaaS
Which cloud type offers maximum security and data control?	Private cloud
Which cloud type is most cost-effective and scalable?	Public cloud
Which provider is the industry leader with the most extensive offerings?	AWS
What does EC2 stand for?	Elastic Compute Cloud
What does the instance type (t2, t3, m5) describe?	How powerful the virtual computer is

7. Cloud Security Implications

The Shared Responsibility Model

Cloud security is a partnership between the provider and the customer. The cloud provider secures 'the cloud' (physical hardware, network infrastructure, and hypervisors). The customer secures 'what is in the cloud' (data, access controls, OS configuration, application security). The exact boundary depends on the service model: in IaaS the customer owns more; in SaaS the provider owns more.

Responsibility	Provider Always Owns	Customer Always Owns
Physical security	Data centres, physical access control, hardware replacement	—
Network edge	DDoS protection, network equipment, backbone security	—
Virtualisation	Hypervisor security, VM isolation	—
Identity & Access	—	User accounts, IAM roles, MFA, password policies
Data encryption	Encryption-at-rest tools available	Decision to encrypt, key management

Application code	—	Secure coding, dependency patching, input validation
Compliance	Compliance certifications (ISO, SOC2, PCI-DSS)	Compliance of workloads built on the platform

Top Cloud Security Risks

- Misconfigured storage — open S3 buckets, public Azure Blob containers exposing sensitive data. One of the most common cloud breach causes.
- Overly permissive IAM roles — 'least privilege' violations where users or services have more permissions than needed.
- Exposed credentials — AWS access keys committed to public GitHub repositories are scraped by bots within minutes.
- Insecure security groups — EC2 instances with port 22 (SSH) or port 3389 (RDP) open to 0.0.0.0/0 (the entire internet).
- Public EC2 instances with no patching — cloud instances are internet-facing and must be kept patched and hardened.
- Insufficient logging — CloudTrail/Azure Monitor/GCP Logging must be enabled to detect unauthorised activity.

8. Key Takeaways & Glossary

8.1 Key Takeaways

- Cloud computing delivers computing services over the internet — pay as you go, no upfront hardware investment.
- Cloud is built on virtualisation, containers, networking, and storage — all the fundamentals you have studied.
- Scalability = handling unexpected increases in traffic — the key cloud advantage over on-premises infrastructure.
- IaaS: rent infrastructure (EC2). PaaS: rent a platform (Elastic Beanstalk). SaaS: rent finished software (Gmail).
- IaaS gives you maximum control but maximum responsibility. SaaS gives you minimum control but minimum responsibility.
- The three deployment models: Public (shared, scalable, cost-effective), Private (dedicated, secure, expensive), Hybrid (best of both, complex).
- AWS is the industry leader. Azure leads in enterprise/hybrid. GCP leads in data analytics and AI.
- EC2 = Amazon's virtual machines in the cloud. Instance type defines CPU/RAM. AMI is the VM template.
- IaaS is used for security labs because it provides full OS access — needed to install tools and simulate attacks.

- The Shared Responsibility Model: provider secures the cloud; customer secures what is in the cloud.

8.2 Core Glossary

Term	Definition
Cloud Computing	Delivery of computing services over the internet on a pay-as-you-go basis.
Scalability	The ability to increase (scale out) or decrease (scale in) computing resources based on demand.
IaaS	Infrastructure as a Service — rent raw computing resources (VMs, storage, networking). Customer manages OS up.
PaaS	Platform as a Service — rent a managed platform to deploy apps on. Customer manages app and data only.
SaaS	Software as a Service — use finished software over the internet. Provider manages everything.
Public Cloud	Shared cloud infrastructure owned by a provider (AWS, Azure, GCP) — most cost-effective and scalable.
Private Cloud	Dedicated cloud infrastructure for one organisation — maximum security and compliance control.
Hybrid Cloud	Combination of public and private cloud — balance of control, cost, and flexibility.
AWS	Amazon Web Services — the industry-leading cloud provider with the most extensive service catalogue.
Azure	Microsoft Azure — strong in enterprise and hybrid cloud, deep Microsoft ecosystem integration.
GCP	Google Cloud Platform — known for data analytics, AI/ML tools, and Kubernetes (GKE).
EC2	Elastic Compute Cloud — Amazon's virtual machines in the cloud.
Instance	A single running EC2 VM. You add/remove instances to scale your application.
Instance Type	Defines the performance (vCPU and RAM) of an EC2 instance. Examples: t2.micro, m5.large.
AMI	Amazon Machine Image — the OS and software template used to launch EC2 instances.
Region	A geographic location of AWS data centres. Choosing close regions reduces latency for users.
Security Group	AWS virtual firewall — controls inbound/outbound traffic rules for EC2 instances.

Shared Responsibility	Cloud security model: provider secures the cloud infrastructure; customer secures their data and applications.
Cloud Bursting	Hybrid cloud feature — overflow workloads move to public cloud when private capacity is exhausted.
S3	Amazon Simple Storage Service — object storage for files, backups, and static website hosting.

8.3 Next Steps on TryHackMe

- **Immediate:** Continue with the Operating Systems Introduction room — how OS components manage the VMs and cloud services covered here.
- **Security:** Study the Cloud Security Fundamentals room — security controls, misconfigurations, and attack vectors in cloud environments.
- **Cloud Security:** Complete the AWS rooms (S3, IAM, EC2 Security) — hands-on cloud security from an offensive and defensive perspective.
- **Offensive:** Study the Introduction to Cloud Security Attacks — how attackers exploit misconfigured cloud resources.
- **Certification:** Work towards AWS Cloud Practitioner or CompTIA Cloud+ certification for formal cloud credentials.

Cloud Is Now the Default

The majority of new applications are built on cloud infrastructure. TryHackMe runs on cloud. Every major tech company uses cloud. Every aspiring security professional will encounter cloud environments — as targets, as defensive infrastructure, or as the platform their security tools run on. The fundamentals covered in this room are not optional background — they are core skills for every security career path.

Source: TryHackMe — Cloud Computing Fundamentals Room | tryhackme.com/room/cloudcomputingfundamentals